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The critical importance of stack pressure in batteries

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Stack pressure is essential for ensuring solid–solid interfacial contact in batteries, where it plays a particularly crucial role. The wide variation in stack pressure levels in batteries has perplexed researchers seeking to determine an optimal value and has led to substantial gaps in our understanding of how stack pressure interacts with battery chemistry. Here we highlight the critical importance of stack pressure in batteries and advocate for considering a critical stack pressure empirical model as a means to determine the optimal stack pressure. We begin by analysing the broad range of stack pressures, which span multiple orders of magnitude. We then categorize their effects into four distinct stages and explain their roles. Additionally, we examine the electrochemomechanical relationship of stack pressure to reveal a coupled effect. Future research on stack pressure should focus on areas such as benchmarking, diagnosis, spatial distribution and minimization. A deeper understanding of stack pressure will facilitate the development of more reliable and practical battery designs across various chemistries.

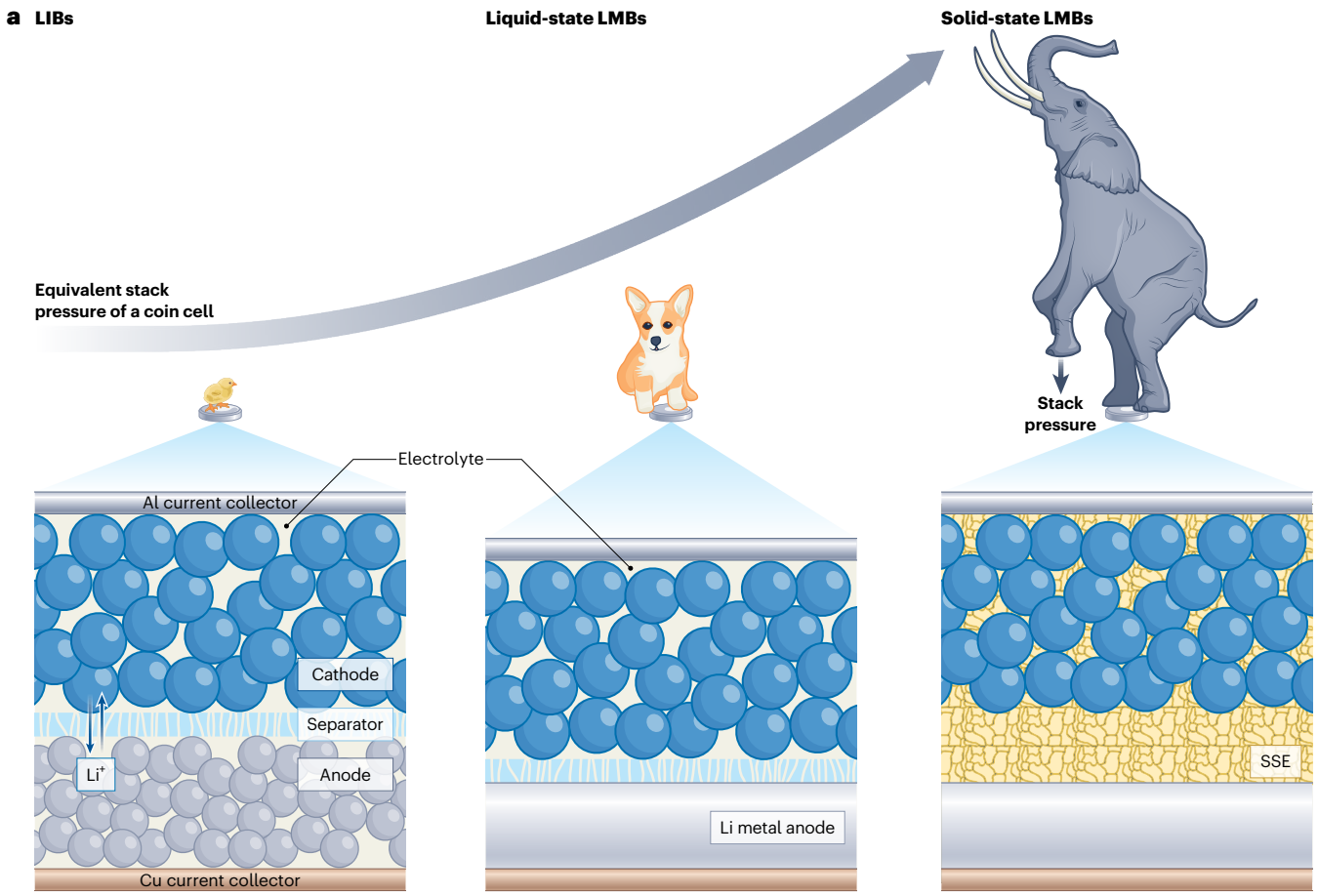
Pursuing batteries that approach 100% Coulombic efficiency is a major target for high-energy-density lithium batteries. This requires comprehensive optimization across various factors, including composition (of the electrolyte and substrate, for example), structure (for example, use of a three-dimensional host and a gradient host) and operating conditions (such as current density, temperate and stack pressure)[1,2]. Among these, the operating stack pressure emerges as a more critical factor than was previously recognized. On the one hand, elevating the stack pressure alone can significantly enhance the Coulombic efficiency, increasing it from 60 to 90% and prolonging the cycle life by more than six times[3,4]. On the other hand, current high-Coulombic-efficiency (>99%) battery designs are often achieved under varying stack pressures[5]. It is evident that fundamental understanding of stack pressure in batteries remains unclear.

Generally, stack pressure plays an essential role in enhancing solid–solid interfacial contact in batteries[6,7]. Recently, the effects of stack pressure on battery performance have been increasingly recognized. However, the spectrum of stack pressure levels varies significantly across different lithium batteries, spanning multiple orders of magnitude. Even within the same system, the range of stack pressures can differ substantially, especially in solid-state batteries (SSBs). As depicted in Fig. 1a, the typical equivalent stack pressure on coin-sized cells varies widely—from that of a chicken on lithium-ion batteries (LIBs) to that of a dog on liquid-state lithium-metal batteries (LMBs) and even to that of an elephant on solid-state LMBs. Low stack pressures of <50 psi (<0.35 MPa) in LIBs are adequate to ensure good contact between the separator and electrodes. Moderate stack pressures of 0.1 to a few MPa in liquid-state LMBs could help to suppress dendrite growth and fluffy deposition. In contrast, much higher stack pressures, ranging from several to hundreds of MPa in solid-state LMBs, are necessary to prevent layer delamination, severe volume expansion and interfacial contact issues (Fig. 1b). Importantly, although fabrication pressures for solid-state electrolyte (SSE) pelletizing are usually high (~100–500 MPa), this Perspective specifically focuses on operating stack pressures[2].

The MPa-level stack pressure of the battery will be amplified in real applications. For example, in a future electric vehicle powered by approximately 8,000 SSBs, the implementation of a pressurization

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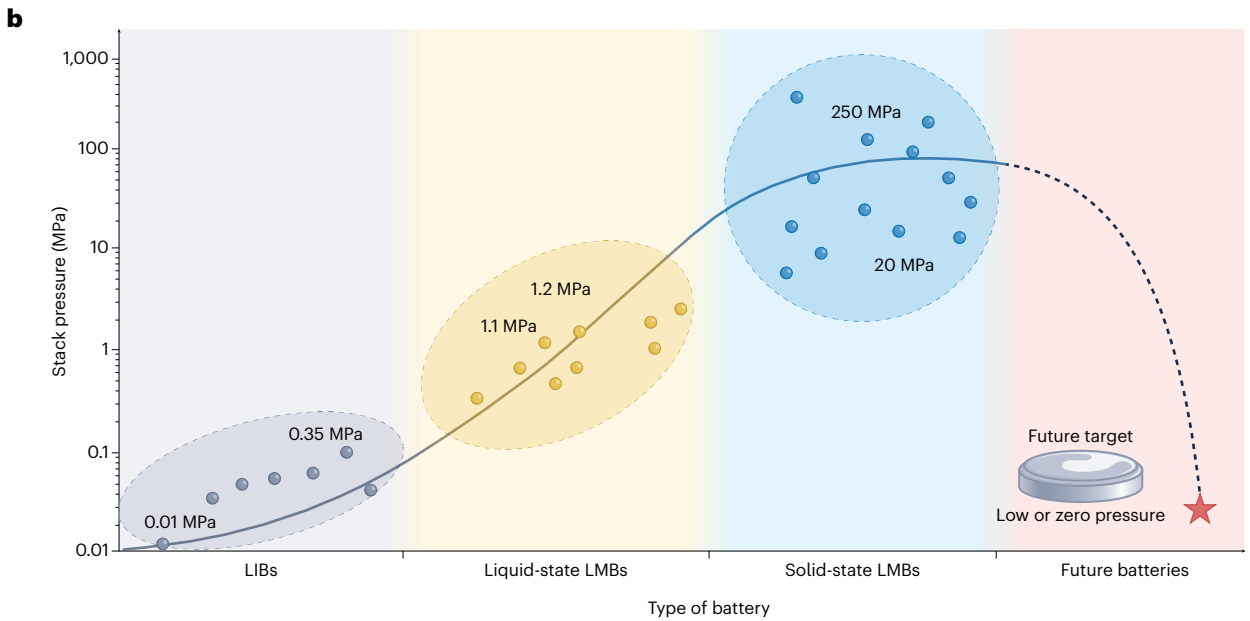


Fig. 1 | Schematic and range distribution of stack pressure in LIBs, liquid-state LMBs and solid-state LMBs. a, Schematic of equivalent stack pressures on coin-sized cells across different kinds of lithium battery. **b**, Range distribution of stack pressures in lithium batteries, showing variation by multiple orders of magnitude. The grey, yellow and blue data points represent pressure values applied to LIBs, liquid-state LMBs and solid-state LMBs, respectively. All data points are included in the Source Data Fig. 1. The solid line illustrates the trend in operating pressure across these battery types, whereas the dashed line with a star indicates the future target for achieving low or zero pressure.

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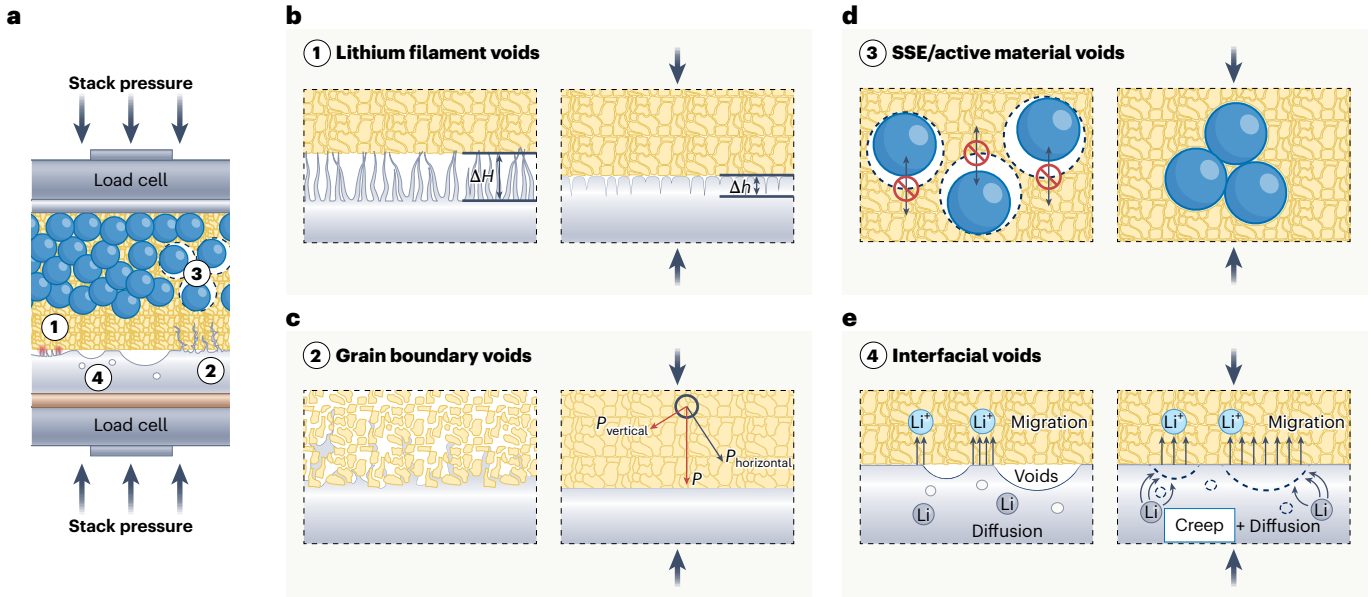


Fig. 2 | Stack pressure effects on preventing voids in batteries. a, Schematic of stack pressure effects on solid-state LMBs. The labels 1–4 relate to panels **b**–**e**, respectively. **b**–**e**, Schematics showing how stack pressure suppresses lithium filament voids (**b**), grain boundary voids (**c**), SSE/active material voids (**d**) and interfacial voids (**e**). ΔH and Δh denote the thickness change of lithium deposition without and with applied stack pressure, respectively, where $\Delta H > \Delta h$. Solid lines depict material outlines, whereas dashed lines represent the original void boundaries without stack pressure.

system poses a major challenge, and ensuring material tolerance under such a high stack pressure level presents another. Battery materials suffer from mechanical fatigue under excessive stack pressure, resulting in a drastic degradation of performance. Besides, the qualitative and quantitative relationships between stack pressure and electrochemical behaviour are ambiguous. Currently, it is challenging to decouple the stack pressure effect independently, partially due to the electrochemomechanical relationship of stack pressure.

These uncertainties surrounding stack pressure not only perplex researchers attempting to determine an optimal stack pressure, but also create considerable gaps in our understanding of the interplay between chemistry and stack pressure. In this Perspective, we discuss the critical roles of stack pressure in batteries by analysing its effects across varying levels. Our analysis indicates that electrochemical performance peaks are achieved within a specific stack pressure range. We therefore propose a critical stack pressure (CSP) empirical model as a benchmark for standardizing battery evaluation. We emphasize that the major contribution of stack pressure is in mitigating void formation during the initial stage. Moreover, we identify a threshold at which the beneficial effects of stack pressure are maximized, beyond which parasitic effects begin to dominate. We also discuss the electrochemomechanical relationship and spatial distribution effects of stack pressure. Looking forward, we advocate for focused research on this fundamental aspect, to guide further development of better batteries.

Suppression of voids

Void formation is one of the main culprits of battery degradation and failure, frequently arising from material volume changes, crack propagation or irregular lithium plating/stripping. Stack pressure can mitigate the formation of voids, including in lithium filaments, grain boundaries, SSE/active materials and interfaces, during cycling (Fig. 2a). These effects can be attributed to enhanced atomic mobility and solid–solid contact under mechanical compression. Consequently, voids or cracks can be effectively alleviated through the spontaneous evolution by localized internal stress originating from stack pressure[8]. Even though the impact of local stress is more severe in commercial electrodes (such as graphite or silicon anodes), the effects of stack pressure are ubiquitous across battery chemistries and are particularly profound in SSBs, as exemplified in solid-state LMBs (Fig. 2).

The presence of loose, mossy and dendritic lithium filaments on the anode side results in numerous voids and a large surface area, compromising battery efficiency. Typically, relatively high pressures are preferred, to eliminate lithium filament voids and regulate lithium deposition behaviour, as illustrated in Fig. 2b. The mechanical suppression of dendrites was first theoretically explored by Monroe and Newman[9], who showed that, when the shear modulus of the SSE or polymer separator exceeds twice that of lithium metal (G_{Li} = 4.2 GPa), the growth of dendritic lithium can be successfully inhibited. Wang and co-workers[10] utilized atomic force microscopy cantilever pressure to generate local stack pressure (~13 MPa) on a single lithium whisker, observing suppression of axial growth and lateral expansion into lithium particles. Meng and co-workers[11] found that stack pressure directs lithium growth at the microscale by adjusting surface energy and densifies lithium deposits at the nanoscale through mechanical constraints (Fig. 2b). In liquid-state LMBs, a modest stack pressure of 0.35 MPa enabled highly dense lithium (99.49% of the theoretical density) deposition[11]. In contrast, SSBs typically require at least 20 MPa to regulate deposited lithium porosity and reduce the formation of inactive lithium[12,13].

Grain boundary voids in most SSEs represent critical practical challenges not addressed by Monroe and Newman's theoretical framework. Stack pressure can facilitate lithium dendrite growth across SSEs along the grain boundary and cause potential cell shorting, even at very low current densities[14,15]. One potential approach to mitigating this issue is to cycle at high stack pressures (Fig. 2c), during which the stack force at the local grain boundary can be divided into two component forces. The force perpendicular to the interface direction can suppress the penetration of lithium dendrites. However, due to the irregular SSE particles, the component force perpendicular to the interface direction varies, resulting in uneven suppression across different grain boundaries. High stack pressures above 10 MPa are generally considered necessary to achieve sufficient suppression, yet such high values limit practical viability[16]. Alternative strategies, such as applying lateral pressure

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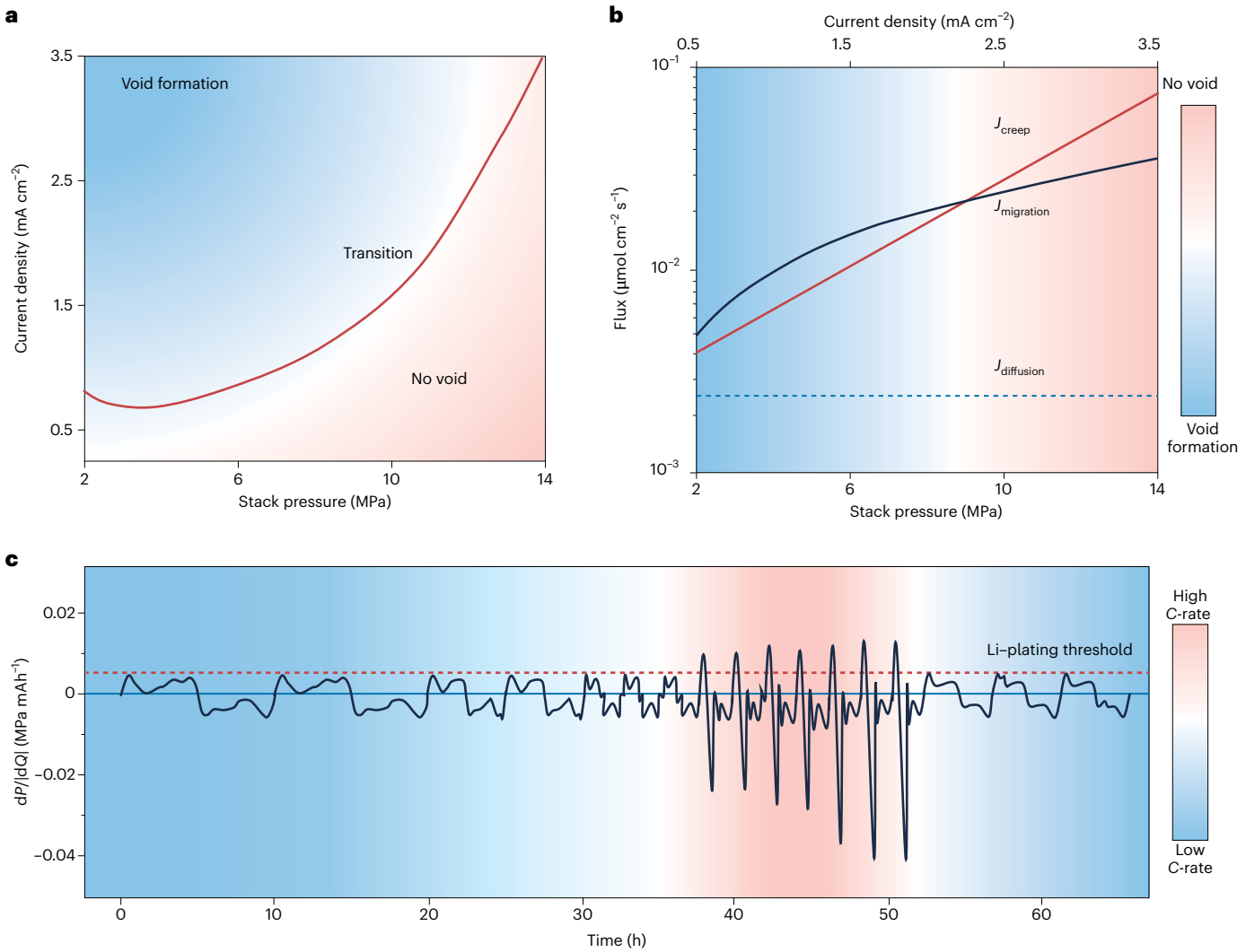


Fig. 3 | Electrochemomechanical relationship of stack pressure in batteries, where electrochemical behaviour and stack pressure serve as two distinct dimensions. a, Map of interfacial void formation as a function of stack pressure and current density in batteries. The transition curve represents the critical boundary. **b**, Lithium flux as a function of lithium atom creep, lithium atom diffusion and lithium-ion migration with stack pressure and current density. **c**, The operando d*P*/d*Q* signal serves as an indicator with which to detect lithium plating events, exemplifying a typical case that combines stack pressure and electrochemical behaviour. The C-rate is defined as the ratio of charge or discharge current to the battery's theoretical capacity. The solid blue line indicates d*P*/d*Q* = 0, whereas the red dashed line is the lithium-plating threshold defined by the maximum of d*P*/d*Q* at 0.2 C. Panels adapted with permission from: **a**,**b**, ref. 21, Wiley; **c**, ref. 32 under a Creative Commons license CC BY 4.0.

and creating single-crystal SSEs, could alleviate grain boundary void formation and concurrently provide better suppression of lithium growth along grain boundaries.

Another contributor to void formation is the detachment of active materials from the SSE in electrodes. Detachment commonly arises from factors such as mechanical stress, thermal expansion/contraction, chemical reactions or structural shifts within the battery system, creating gaps and cutting off the lithium transport pathway[17]. Stack pressure minimizes this effect, thereby preventing disengagement between the SSE and active materials in electrodes (Fig. 2d). Therefore, the application of an appropriate stack pressure is crucial to maintain battery stability and longevity, by averting the occurrence of detachment-related issues.

SSE/electrode interfacial voids are another critical degradation mechanism, gradually weakening interfacial contact and accelerating failure[18]. Under applied pressure, microscopic lithium creep can be triggered through dislocation climb, building on the original lithium diffusion at the anode (Fig. 2e)[19,20]. The combined fluxes of lithium atoms due to creep and diffusion surpass lithium-ion migration, as expressed in equation (1)[21]:

$$J_{creep} + J_{diffusion} > J_{migration} \quad (1)$$

where *J* represents the magnitude of flux. This causes vacancies to transport away from the interface, thus replenishing voids at the interface. As this replenishment proceeds, localized stress at the SSE/electrode interfacial contact point can be effectively relieved, leading to uniform Li^+ ion migration during the stripping process, ultimately yielding a flat interface.

Electrochemomechanical relationship of stack pressure

The influence of stack pressure on battery performance suggests a potential link to electrochemical behaviour, although the underlying mechanism remains unclear[22]. As depicted in Fig. 3, electrochemical behaviour and stack pressure can be considered as two distinct yet

interacting dimensions, reminiscent of the classical coupling between electricity and magnetism. A deep understanding of this interaction will be useful for uncovering the failure mechanisms of anodes and cathodes under specific pressure conditions. Currently, many insights into this rely heavily on simulations, due to the lack of definitive experimental data.

Stack pressure can mirror electrochemical behaviour as it changes periodically with the electrochemical process[23]. This phenomenon offers a preliminary basis for differentiating between normal and harmful states, including interfacial evolution[24], short circuiting[25] and gas evolution[26]. Chen and co-workers[21] proposed a function linking stack pressure to current density as a means to assess void formation in solid-state LMBs (Fig. 3a). This relationship is established based on the competition among creep, diffusion and migration (in the SSE), as influenced by stack pressure (Fig. 3b)[21]. Zhang and co-workers[27] further proposed a threshold condition based on the Bond equation (stack pressure (P)/current density (I) > 25 MPa cm^{-2} mA^{-1}) to assess the mechanical stability at the SSE/cathode interface. Xiao and co-workers[28] examined Li^+ ion detour behaviour by monitoring the spatial distribution of stack pressure across the battery surface to reduce swelling. Despite these advancements, accurately capturing the onset of harmful events before they lead to failure is proven to be a challenging task.

Furthermore, a comprehensive understanding of the pressure and electrochemical relationship across the entire battery process is still lacking. For example, in the stripping process the critical current density is theoretically positively correlated with increasing stack pressure, whereas in the plating process a negative correlation is observed[29,30]. Deeper understanding of different processes is therefore needed. Compounding the complexity, the accumulation of residual solid electrolyte interphase and isolated lithium would gradually increase the baseline stack pressure, but this non-routine variation cannot be accurately correlated with specific electrochemical reactions[31].

To address these issues, we established a relationship between lithium plating and lithium intercalation by defining a specific threshold based on the change in stack pressure per unit charge (dP/dQ) in LIBs, as shown in Fig. 3c (ref. 32). The corresponding relationship between stack pressure and electrochemical behaviour was captured in real time. In general, as the charging capacity increases, so too does the operando pressure. However, once lithium deposits on the graphite surface, it further re-intercalates into bulk graphite particles. This re-intercalation enhances battery capacity while concurrently decreasing battery stack pressure, leading to an asynchrony in reaching the peak stack pressure and charge capacity. Elucidating these behaviours remains system specific, highlighting the need for further refinement in the development of universal descriptors and reliable benchmarking methods.

Excessive stack pressure

Stack pressure is a double-edged sword, with an optimal range for effective action in batteries. Once this threshold is surpassed, escalating stack pressure can be counterproductive, mainly including undesirable material degradation and shorting. When the stress from stack pressure surpasses the yield strength, battery materials become unable to withstand such immense stress, leading to plastic deformation or even structural damage. These deleterious repercussions stem from the finite pressure tolerance inherent to battery materials. We analysed the issues and failure mechanisms of liquid- and solid-state LMBs associated with excessive stack pressure (Fig. 4), with a particular focus on elucidating the common challenges of each type.

As the stack pressure applied to the liquid- and solid-state LMBs escalates, the current collector is more susceptible to deformation and preferential rupture (Fig. 4c), primarily due to its lower deformability (~2%) compared with that of the separator (70–150%), leading to electrode failure[33]. The escalation of stack pressure primarily induces electrode cracking (Fig. 4d,e) and material pulverization (Fig. 4f,g)[34], resulting in poor interfacial contact, material deactivation and other associated issues. In contrast, the lithium anode exhibits a distinctive response under constraint due to its soft and creep characteristics, distinguishing it from other materials. Excessive stack pressure exacerbates lithium creep and plastic deformation, resulting in lithium being forced out through gaps (Fig. 4h,i)[35]. This can also promote dendritic growth across the micropores of the separator and SSE pellet (Fig. 4j,k), leading to short circuit[36].

Considering the disparities between separators and SSEs, the reactions of liquid- and solid-state LMBs under excessive stack pressure vary. In liquid-state cases, excessive pressure may lead to the closure of separator pores, hindering ion transport and elevating impedance (Fig. 4l)[37]. This could drastically affect battery performance, especially during extremely fast charging scenarios[38]. In solid-state cases, fluctuations in stack pressure at the MPa level can induce mechanical fatigue in the SSE, leading to crack formation as the material attempts to relieve internal stress (Fig. 4m)[39]. Over time, the SSE may fracture into multiple segments, impeding Li^+ ion migration and creating voids conducive to the growth of lithium filaments.

Generally, insufficient stack pressure fails to ensure optimal interfacial contacts, whereas excessively high values can pose potential hazards. Between these extremes lies a critical range where the benefits of stack pressure are maximized, striking a balance between electrochemical behaviour and stack pressure utilization. Hence, it is crucial to promptly investigate and determine this pivotal stack pressure range.

CSP

The minimization of voids necessitates adequate stack pressure, yet determining the optimal level remains unclear. Although many strategies have shown increased Coulombic efficiency in various battery systems, their correlation with stack pressure remains unknown due to the lack of stack pressure recording. It has been found that the interfacial resistance decreases to a certain extent and remains stable with increasing stack pressure, and other factors have similar variation trends with pressure[40,41]. This indicates that stack pressure does not exhibit a linear correlation with Coulombic efficiency, where there might exist a threshold instead[42].

We summarize the statistics of Coulombic efficiency (Fig. 5a) and stack pressure (Fig. 5b) in battery systems, including lithium-, sodium-, potassium-, zinc-, magnesium-, calcium- and aluminium-based batteries. Currently, the consideration of stack pressure is primarily focused on lithium-based batteries; its relevance in other systems remains unclear. In lithium-based batteries, we have listed the battery Coulombic efficiency along with the stack pressure. However, for other batteries, most lack stack pressure information due to insufficient data availability. Figure 5a also depicts a localized amplification in lithium- and sodium-based batteries (cases with and without stack pressure information). Clearly, the stack pressure in other battery types (for example, potassium-, zinc-, magnesium-, calcium- and aluminium-based batteries) needs to be explored in the future. We advocate for a more focused exploration of this fundamental aspect across all types of battery.

Providing a comprehensive and global perspective on stack pressure effect would greatly benefit the battery community. Based on our statistical analysis, we propose a CSP empirical model, as depicted in Fig. 5c, that illustrates the general relationship between Coulombic efficiency and stack pressure in liquid- and solid-state LMBs. In liquid-state LMBs (blue curve in Fig. 5c), the initial stage (L1) of Coulombic efficiency variation demonstrates a noticeable increase as the stack pressure rises to a critical point (termed CSP_L) at which the Coulombic efficiency reaches a maximum level. During this process, void elimination initiates and propagates, substantially reducing resistance and promoting uniform ionic transportation. In the subsequent stage (L2), the Coulombic efficiency reaches a plateau and remains relatively stable as the stack pressure continues to rise. On the one hand,

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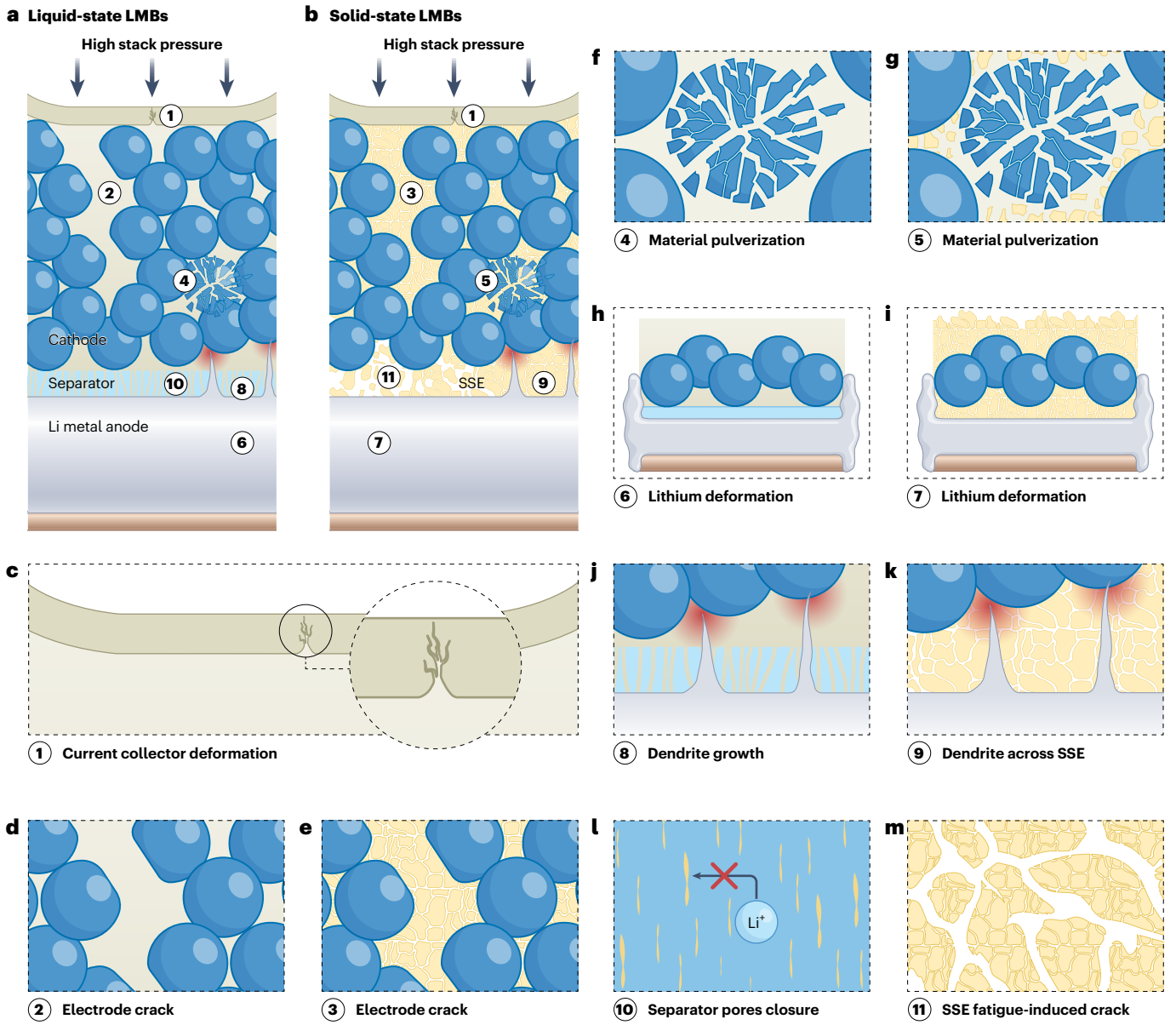


Fig. 4 | Issues and failure mechanisms posed by excessive stack pressure. **a**,**b**, Schematics of excessive stack pressure on liquid- (**a**) and solid-state LMBs (**b**). The labels 1–11 relate to panels **c**–**m**, respectively. **c**, Current collector deformation or rupture under excessive stack pressure, leading to layer lamination and threatening battery seamlessness. **d**–**i**, Electrode crack (**d** and **e**), material pulverization (**f** and **g**) and lithium deformation (**h** and **i**) in liquid- (**d**, **f** and **h**) and solid-state LMBs (**e**, **g** and **i**). **j**,**k**, Lithium dendrite overgrowth across separator pores (**j**) and SSE grains (**k**) towards the cathode side, leading to shorting. **l**, Excessive stack pressure forces the closure of separator pores in liquid-state LMBs. **m**, The SSE crack phenomenon with continued pressure fluctuation induces mechanical fatigue.

further elimination of voids by increasing stack pressure is debatable. On the other hand, any additional increase in Coulombic efficiency might be influenced by other factors. Following this, a failure stage (L3) in battery materials occurs, as discussed in the previous section on excessive stack pressure.

The correlation curve between Coulombic efficiency and stack pressure from stage S1 to stage S3 in solid-state LMBs behaves similarly to that of liquid-state LMBs (orange curve in Fig. 5c). However, solid-state LMBs show an additional stage (stage S0) with the requirement for considerably higher initial stack pressure (termed CSP_{S1}) to initiate battery operation, mainly due to the more notable interfacial challenges. The continuous increase of stack pressure (stage S1) further keeps establishing better interfacial contact, resulting in an increase in Coulombic efficiency until a maximum level (termed CSP_{S2}) is reached[13,43]. The spectrum of this stage in solid-state LMBs is much broader compared with that in liquid-state LMBs. Therefore, the CSP empirical model in solid-state LMBs exhibits two CSP thresholds: CSP_{S1} and CSP_{S2}.

Based on this empirical model, we explore the failure sequence of battery materials under excessive pressure (as shown by the dashed lines in Fig. 5c). Exceeding the stable plateau could trigger a sequence of material degradations, leading to battery failure. The failure process (such as lithium deformation, current collector fracture, electrode pulverization, SSE breakdown, separator failure and so on) warrants further exploration.

The CSP empirical model provides evidence to define the applied stack pressure and assists researchers in quantifying the optimal level. Although current research offers limited examples of

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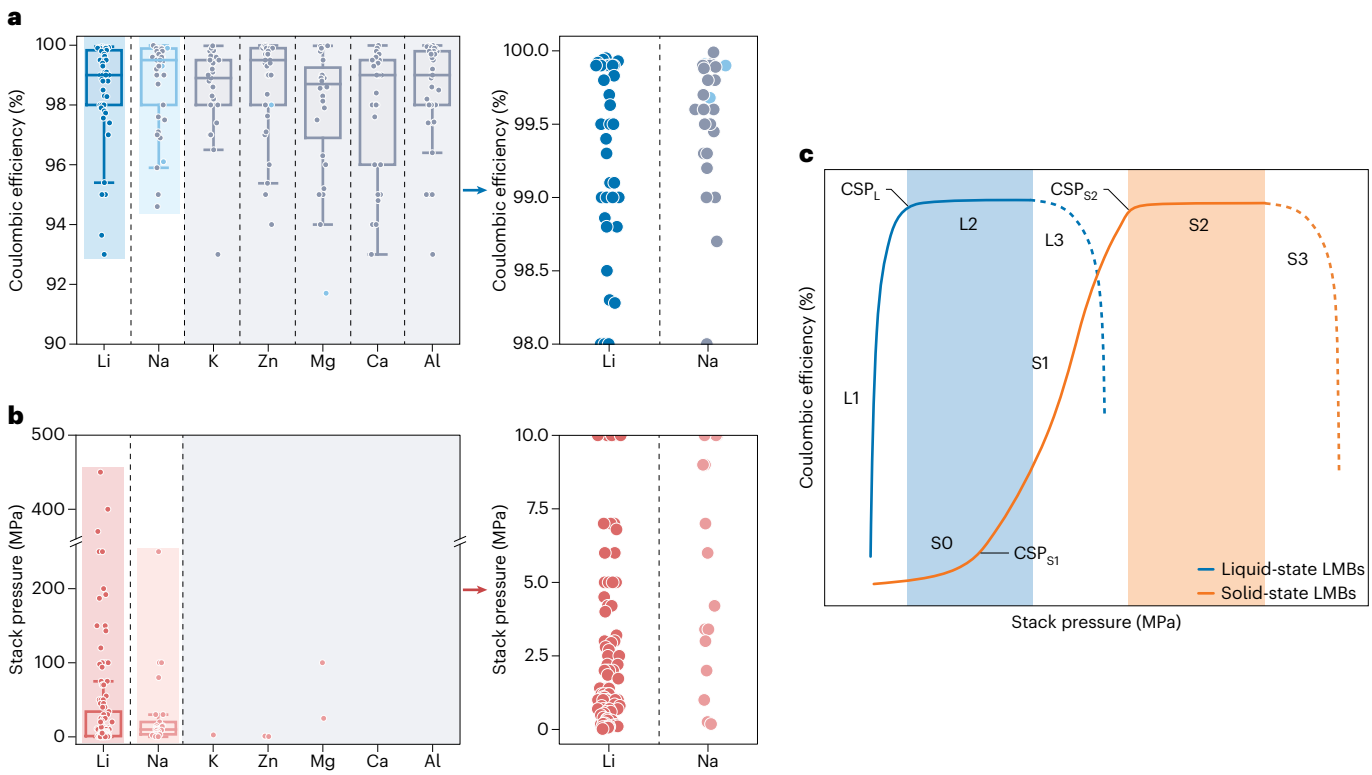


Fig. 5 | CSP empirical model and its analysis in batteries. a, Statistic Coulombic efficiency (left) and its amplification (right) across different types of battery. All data represent measured Coulombic efficiencies in batteries, as reported in published articles (summarized in Source Data Fig. 5), where blue points denote cases with stack pressure information and grey points represent those without. **b**, Statistic stack pressure (left) and its amplification (right) across different types of battery. All data represent experimentally applied stack pressures reported in published articles. The lack of statistical data on stack pressure for potassium-, zinc-, magnesium-, calcium- and aluminium-based batteries is primarily because of limited studies in this field, which warrants more exploration. In **a** and **b**, the box plots display median values (central lines) and 25–75% limits (box edges) and the whiskers extend to 1.5× the interquartile range. **c**, Relationship between Coulombic efficiency and stack pressure in liquid- and solid-state LMBs. Optimal stack pressure ranges are indicated by the shaded regions.

CSP empirical model validation, several studies have demonstrated promising trends[44,45]. Specifically, the optimal stack pressure range corresponds to the blue and orange zones marked in Fig. 5c. CSP_L and CSP_{S2} can be used as important guidelines for battery design principles as they indicate the achievement of high Coulombic efficiency but the requirement for reduced stack pressure. However, determining the CSP for each type of battery requires systematic studies because this value is system specific. We propose a potential method for its measurement. By continuously measuring the Coulombic efficiency values of batteries while gradually increasing the stack pressure, the minimum stack pressure that yields the highest Coulombic efficiency can be identified as the CSP empirical model for a given battery system. In addition to fixed stack pressure, dynamic stack pressures (such as 25 MPa SSE/Li interfacial formation pressure before reducing to 5 MPa for cycling)[36] and multiple-step protocols can also be derived from the CSP empirical model.

Stack pressure spatial distribution

Stack pressure typically serves as a global descriptor, but its local effects are often overlooked. The distribution of stack pressure is spatially inhomogeneous across the cell, influenced by factors such as surface roughness, non-uniform lithium deposition/dissolution, dead lithium accumulation and the pressurization device. The best pressure uniformity conditions limit the maximum effectiveness of batteries[46,47], whereas worse pressure uniformity conditions make them susceptible to triggering local side reactions, including dendrite growth, overcharging and thermal runaway.

The essential requirement for achieving ideal stack pressure application lies in the necessity for conformal, constant, spatially uniform and self-adaptive pressure apparatuses. Based on these characteristics, pressure-responsive mechanical devices (pneumatic actuators), volume-absorbable materials or systems (airbag holders, silicone foams and compression springs)[36,38,48] and fluid pressure apparatuses (air fluid isostatic pressure holders)[49] have been proven to effectively alleviate the spatial variation of stack pressure. Additionally, the integration of three-dimensional micropatterned SSEs[50], elastic solid electrolytes and responsive interfacial layers can facilitate the contact area among interfaces.

Future directions

Despite the crucial roles of stack pressure in influencing electrochemical behaviour, several key challenges remain unresolved. These include—but are not limited to—the establishment of benchmarking standards, stack pressure diagnosis methods, spatial homogeneity control and stack pressure optimization (Fig. 6).

The first priority is to establish a benchmark for stack pressure. It is imperative to standardize and unify stack pressure protocols for cell benchmarking, as researchers often struggle with the selection of stack pressure due to the wide range reported across different studies[51]. Therefore, we propose the CSP empirical model as a reliable guideline and standard across various battery systems and pressure setting protocols.

The second challenge is diagnosing battery electrochemical states through stack pressure. Currently, the connections between stack pressure and electrochemical behaviour are quite complicated, necessitating multiple analytical techniques to establish reliable intrinsic correlations between them. In LIBs, dP/dQ can serve as a high-precision indicator for monitoring real-time health status. Battery operation

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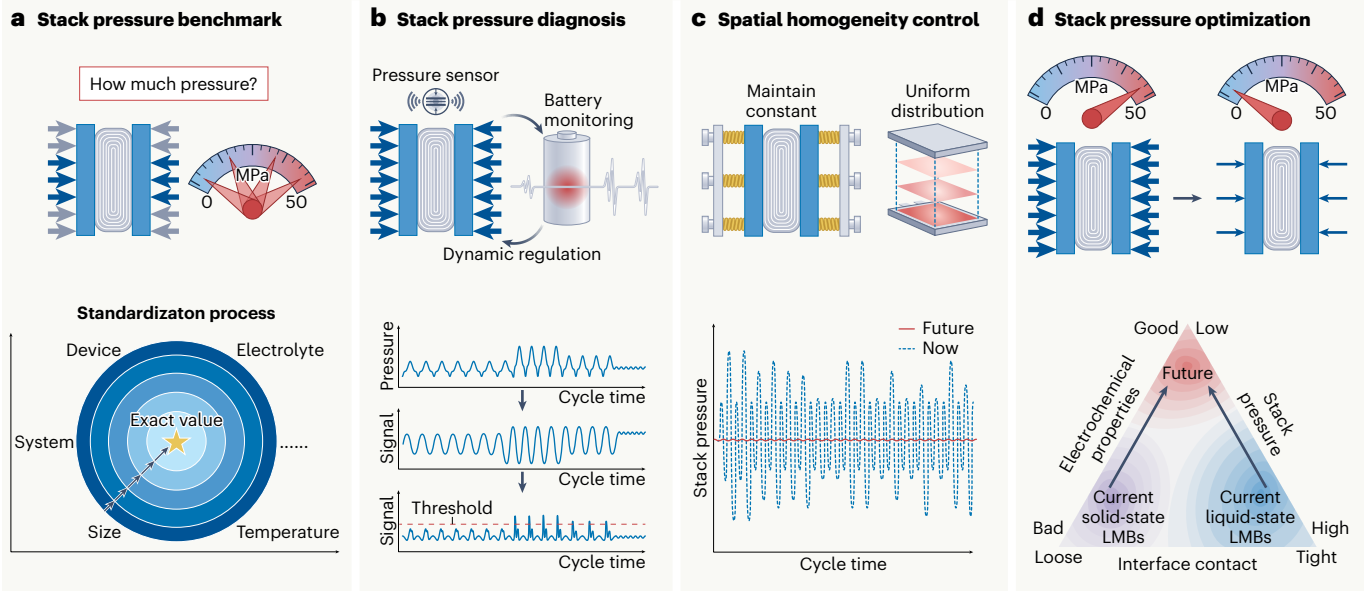


Fig. 6 | Future directions for stack pressure in batteries. a, The establishment of a stack pressure benchmark is critical for determining the optimal stack pressure range for specific battery systems. **b**, Schematic of the diagnosis of battery electrochemical states through dynamic stack pressure variation. **c**, Controlling spatial pressure homogeneity helps to mitigate uneven ageing and prolongs battery lifespan. **d**, Stack pressure optimization is necessary for enabling practical battery applications.

can therefore be dynamically regulated adaptively from the feedback of stack pressure, which provides possibilities for the realization of smart and multi-functional batteries.

The third challenge is achieving spatial homogeneity in stack pressure. Stack pressure is a dynamic quantity that varies temporally and spatially. The essence of uniform pressure design lies in minimizing deviations from the desired applied stack pressure, thus facilitating the spatial distribution of the stack pressure. In equations (2) and (3), we propose a method for measuring this uniformity:

$$\Delta P_{\mathrm{norm}}(x,y) = \frac{P(x,y) - \bar{P}}{P_{\mathrm{max}}(x,y) - P_{\mathrm{min}}(x,y)} \tag{2}$$

$$\sigma^2 = \frac{1}{n}\sum_{i,j=1}^{n}\left(\Delta P_{\mathrm{norm}}(x_i,y_j) - \overline{\Delta P_{\mathrm{norm}}(x,y)}\right)^2 \tag{3}$$

where $\Delta P_{\mathrm{norm}}(x,y)$ represents the normalized differential stack pressure at each point (x,y); where x and y are the row and column of the sensor matrix, respectively) and σ^2 quantifies the discrete degree of stack pressure distribution. A σ^2 value of <3% indicates greater uniformity.

The fourth challenge is optimizing the applied stack pressure. The emerging trend towards reducing or eliminating stack pressure is driven by practical considerations (typically requiring a pressure of <0.1 MPa). Stack pressures up to hundreds of MPa, as seen in many current studies, are impractical due to technical challenges, high manufacturing costs and safety hazards arising from increased mechanical stresses on battery components. Future studies should focus on elucidating the relationship between stack pressure and various failure modes. The development of new technologies that require low or zero stack pressure is expected to become a key focus in the future. Potential approaches to accomplishing this target include: constructing elastic interfaces by incorporating materials capable of flowing or deforming; optimizing electrode materials structurally; exploring stress management techniques; and integrating high-modulus, highly flexible battery components[52]. These approaches can help to offset a substantial portion of the external stack pressure required, leading to a substantial reduction in overall stack pressure.

Conclusion

Considering stack pressure and establishing a benchmark for it are essential for future battery advancements. A key priority is to build robust correlations between electrochemical behaviour and stack pressure. This endeavour will offer great opportunities for understanding the complex electrochemomechanical relationship of stack pressure and uncovering its intricacies. Simultaneously, incorporating the CSP model as a design descriptor is imperative to advancing battery efficiency, leading to the development of better, safer and more reliable batteries. Additionally, considering the material tolerance to stack pressure is important to prevent battery failures. It is hoped that leveraging the opportunities highlighted in this Perspective will inspire meaningful progress in efficient energy storage systems.

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Y.Y. conceived of the project. Y.Y., Q.L. and H.L. analysed the statistical data and wrote the manuscript. K.J.L. helped with the literature studies. All authors contributed to discussion of the results.

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